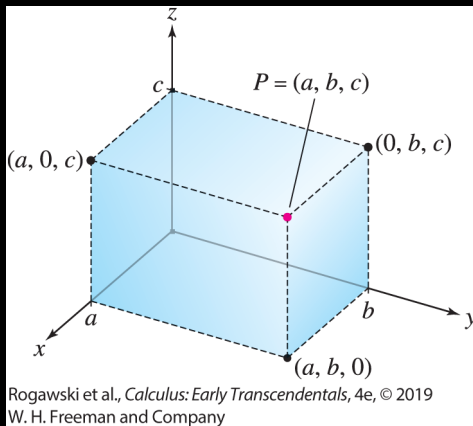


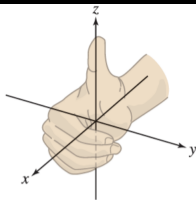
Section 12.2 - Surfaces, Vectors, Curves

MATH 230/231

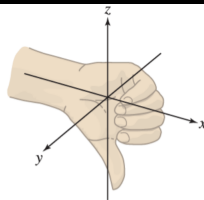
Understanding 3D Space



Right-Hand Rule



(A) This axis labeling satisfies the right-hand rule.



(B) This axis labeling does not satisfy the right-hand rule because the thumb points in the negative z -direction.

Rogawski et al., *Calculus: Early Transcendentals*, 4e, © 2019
W. H. Freeman and Company

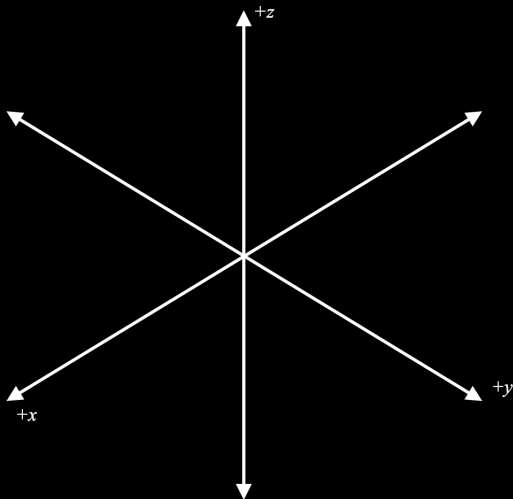
Distance in $\mathbb{R}^3$

Distance in two-dimensions extends as you would expect to three dimensions. If $P = (a_1, b_1, c_1)$ and $Q = (a_2, b_2, c_2)$, then $|P - Q|$ is

$$|P - Q| = \sqrt{(a_2 - a_1)^2 + (b_2 - b_1)^2 + (c_2 - c_1)^2}$$

Planes

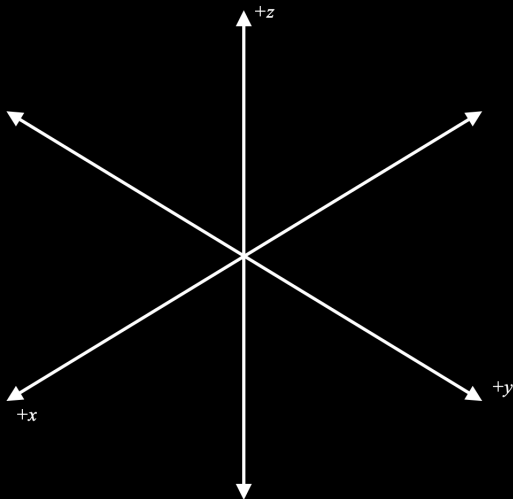
In $\mathbb{R}^3$, $x = a$, $y = b$, and $z = c$ corresponds to a plane that is parallel to the yz -plane.



Cylinders

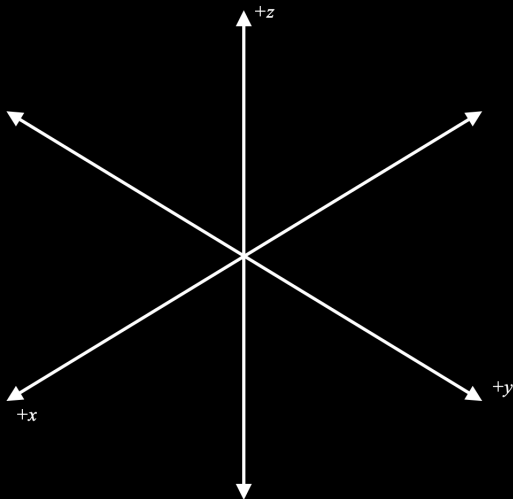
In $\mathbb{R}^2$ the equation $(x - a)^2 + (y - b)^2 = R^2$ corresponds to a **circle**.

In $\mathbb{R}^3$ the same equation, $(x - a)^2 + (y - b)^2 = R^2$ corresponds to a **cylinder**.



Spheres

In $\mathbb{R}^3$ the equation $(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$ corresponds to a **sphere**.

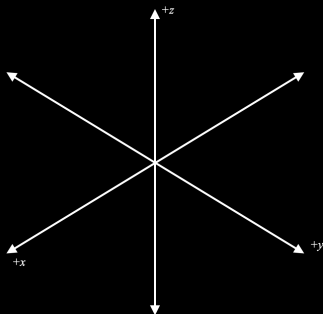
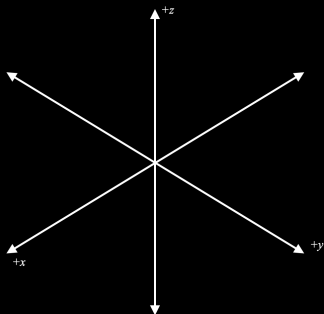


Example

Describe the sets of points defined by the following conditions:

a. $x^2 + y^2 + z^2 = 4, y \geq 0$

b. $(x - 3)^2 + (y - 2)^2 = 1, z \geq -1$

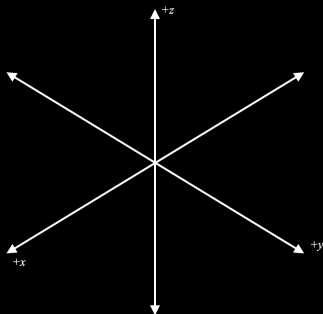
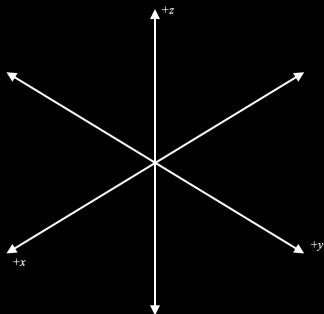


Example

Sketch the following vectors in three dimensions:

a. $\mathbf{v} = \langle 1, 2, 3 \rangle$

b. $\mathbf{w} = \langle 0, 4, 5 \rangle$

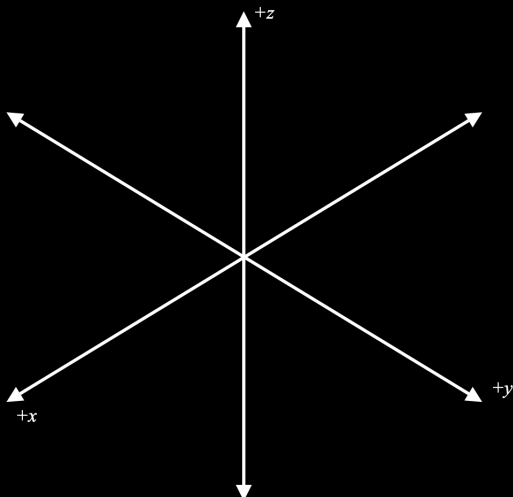


Example

Sketch the vector $\overrightarrow{PQ}$ with the given coordinates and find its magnitude. What are its coordinates?

$$P = (0, 5, 2)$$

$$Q = (4, -2, 6)$$

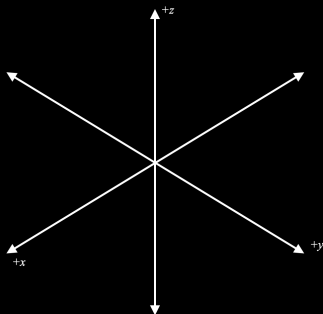
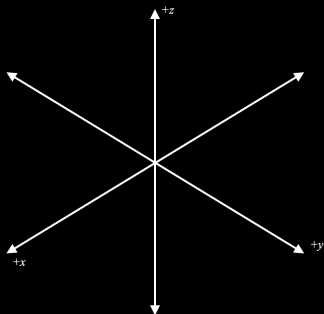


Example

Sketch the **position vector** for each of the following points:

a. $\overrightarrow{OP}$ with $P = (2, 5, -1)$

b. $\overrightarrow{OQ}$ with $Q = (2, 4, 1)$



Some Notes

The usual rules and laws from two dimensions apply in three. That is, we have the following:

- ▶ Two vectors are equivalent if they have the same components.
- ▶ Addition, scalar multiplication are the same.
- ▶ Vectors are parallel if one is a scalar multiple of the other.
- ▶ Vector addition possesses the properties of commutativity, associativity, and the scalar distributive property.

Standard Basis Vectors

The standard basis vectors extend from two dimensions. We now have $\mathbf{k}$ which is the standard basis vector along the z axis.

$$\mathbf{i} = \langle 1, 0, 0 \rangle \quad \mathbf{j} = \langle 0, 1, 0 \rangle \quad \mathbf{k} = \langle 0, 0, 1 \rangle$$

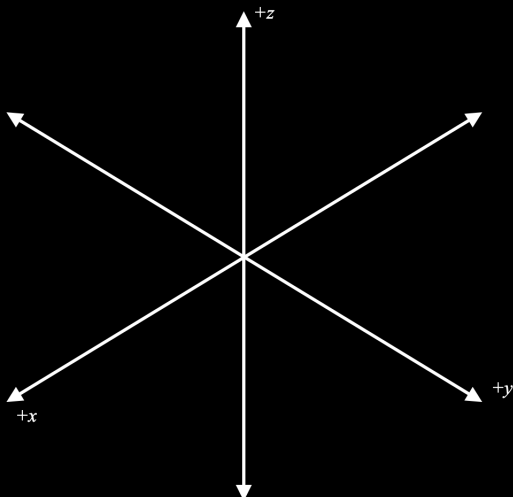
Note that every vector in $\mathbb{R}^3$ is a **linear combination** of these standard basis vectors.

$$\langle a, b, c \rangle = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$$

Example: Write the vector $\langle 9, -4, -8 \rangle$ using the standard basis vectors.

Curves and Lines in $\mathbb{R}^3$

Imagine that I have point that “moves” according to a set of equations, one for each of x , y , and z . That is, the point can be thought of as $P = (x(t), y(t), z(t))$. This point in $\mathbb{R}^3$ defines a **curve** if we instead think of it as a vector starting at the origin, $\mathbf{v} = \langle x(t), y(t), z(t) \rangle$.



Defining Lines in $\mathbb{R}^3$

For a line to be defined in two dimensions, we know that we require two things - a slope and a point. Then we can construct the typical equation of $y = mx + b$. However, in three dimensions, we now need a point and a **direction vector**. The specification of a point is the same, yet now we need some notion of direction emanating from that point. This is why the direction vector is important.

If I have a point $P = (x, y, z)$ on a line $\mathcal{L}$, and I have a vector that starts at point $P_0 = (x_0, y_0, z_0)$. Suppose we are given that the line is parallel to vector $\mathbf{v} = \langle a, b, c \rangle$. Then, we can say the following

$$\overrightarrow{P_0P} =$$

The Equation of a Line in $\mathbb{R}^3$

We can conclude the **equations of a line** in $\mathbb{R}^3$ is defined as follows. If

$r(t) = \langle x(t), y(t), z(t) \rangle$ and $r_0 = \overrightarrow{OP_0} = \langle x_0, y_0, z_0 \rangle$ then the **vector parametrization** of the line is

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}$$

This can also be expressed as

$$\langle x(t), y(t), z(t) \rangle = \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle$$

The **parametric equations of $\mathcal{L}$** are

$$x(t) = x_0 + at$$

$$y(t) = y_0 + bt$$

$$z(t) = z_0 + ct$$

Examples

- ▶ Find the parametrization for the line through $(1, 2, 3)$ and direction vector $\mathbf{v} = \langle 2, 1, 7 \rangle$.
- ▶ Find the parametric equations for the line through the points $P = (1, 0, 2)$ and $Q = (2, 7, 1)$.

Also Review...

Look over Examples 5 and 6 on page 686 of the textbook. They will help in your understanding and work with lines in $\mathbb{R}^3$.